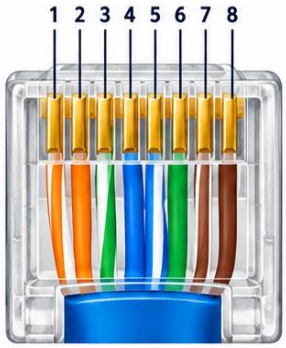
Ethernet standard and specification

Task 1:- Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

RJ-45 Connector Pin Layout



Front view (looking into the connector)

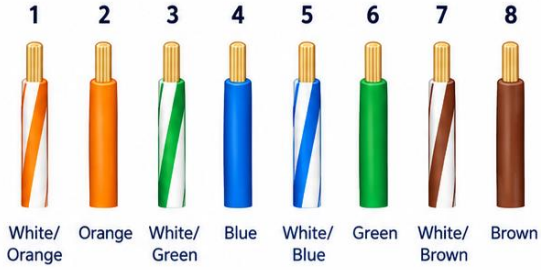
Pin No.	T568A / T568B Color Code
1	White/Orange or White/Green
2	Orange or Green
3	White/Green or White/Orange
4	Blue
5	White/Blue
6	Green or Orange
7	White/Brown
8	Brown

T568A Wiring Standard



T568A order (1 → 8):
White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown

T568B Wiring Standard



T568B order (1 → 8):
White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown

Note: Use the same standard (T568A or T568B) at both ends of the cable for a straight-through connection.

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Task 2:- Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Comparison of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	Cat3 or better twisted-pair cable	Cat5 or better twisted-pair cable	Cat5e or better twisted-pair cable
Maximum cable length	100 meters	100 meters	100 meters
Connector type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Number of wire pairs used	2 pairs	2 pairs	4 pairs
Common application	Older computer networks	Small office and home networks	Modern LANs, gaming, and high-speed internet connections

Task 3:- Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why Think about how transmit and receive pairs are wired in each cable type

Answer: Crossover cable is traditionally used to connect two laptops directly without a switch or router.

1. Why use a crossover cable?

Ethernet devices transmit data through transmit (TX) pairs and receive data through receive (RX) pairs.

- Crossover cable: The transmit and receive pairs are crossed at one end, allowing one laptop's transmit signals to reach the other laptop's receive pins.
- Straight-through cable: The same wiring standard (T568A–T568A or T568B–T568B) is used at both ends, so the wire positions remain consistent.

2. Comparison

Feature	Straight-through cable	Crossover cable
Wiring	Same standard at both ends	T568A at one end, T568B at the other
TX/RX pairs	Same pin arrangement at both ends	TX and RX pairs are crossed
Traditional use	Laptop to switch or router	Laptop to laptop directly
Modern compatibility	May work between laptops	May work between laptops

Task 4:- List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

No.	Device	Likely Ethernet standard	Reason
1	Wi-Fi router	1000BASE-T (Gigabit Ethernet)	Provides high-speed wired connections between the router, computers, and other network devices
2	Desktop computer in a college computer lab	1000BASE-T (Gigabit Ethernet)	Supports fast data transfer, internet access, and sharing files over the college network.
3	Network printer in a college office	100BASE-TX or 1000BASE-T	Connects to the local network so multiple users can print documents reliably.